

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 04



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Recap of Previous Lecture



✓
Topic

GATE PYQ



Topics to be Covered



✓ Topic

Sum of degree theorem

✓ Topic

Degree sequence

✓ Topic

Graphic

✓ Topic

Havel Hakimi's algorithm

✓ Complement of a graph.



Topic : Sum of degree theorem

Handshaking lemma 

✓ Let $G=(V,E)$ be a graph

Where $V = \{v_1, v_2, v_3, \dots, v_{|V|}\}$

then

$$\sum_{i=1}^{|V|} \deg(v_i) = 2 |E|$$

$$\deg(v_1) + \deg(v_2) + \deg(v_3) + \dots + \deg(v_{|V|}) = 2 * |E|$$

Corollary 1:-

If graph $G = (V, E)$ is a directed graph,

then

$$\sum_{i=1}^{|V|} \deg^+(v_i) = \sum_{i=1}^{|V|} \deg^-(v_i) = |E|$$

Corollary 2:-

We know

$$\sum_{i=1}^{|V|} \deg(v_i) = \underbrace{2|E|}$$

it is always even

∴ In any graph G ,
total number of vertices
of 'odd' degree will
always be even

Corollary 3:

In a graph G if degree of Each vertex is exactly ' k ' { i.e. graph is k -regular }

then

$$k \cdot |V| = 2|E|$$

Logic:

$$\deg(v_1) + \deg(v_2) + \dots + \deg(v_{|V|}) = 2|E|$$

$$\underbrace{k + k + \dots + k}_{|V| \text{ times } k} = 2|E|$$

$$\therefore k|V| = 2|E|$$

Corollary 4

In a graph G if degree of each vertex
is at least ' K ' {i.e. greater than or equal to K }

then

$$K|V| \leq 2|E|$$

↳ i.e. $\deg = K$
or

($K + \text{some non-negative integer}$)

Logic:-

$$\deg(v_1) + \deg(v_2) + \dots + \deg(v_{|V|}) = 2|E|$$

$$(K + x_1) + (K + x_2) + \dots + (K + x_{|V|}) = 2|E|$$

$$\text{Where } x_i \geq 0, \forall i$$

Where $x_1 \geq 0$
↑
integer

$$K \cdot |V| + \sum_{i=1}^{|V|} x_i = 2|E|$$

$$\text{↳ } \forall i, x_i \geq 0, \text{ so}$$

$$\sum_{i=1}^{|V|} x_i \geq 0$$

Corollary 5:-

In a graph G if degree of each vertex is at most K

then

$$K|V| \geq 2|E|$$

$$\deg(v_1) + \deg(v_2) + \dots + \deg(v_{|V|}) = 2|E|$$

$$(K - x_1) + (K - x_2) + \dots + (K - x_{|V|}) = 2|E|$$

When $x_i \geq 0, \forall i$

$$K|V| - \left\{ \sum_{i=1}^{|V|} x_i \right\} = 2|E|$$

≥ 0

Note:-

In a graph G ,

degree is at least $\delta(G)$ $\therefore \delta(G) \cdot |V| \leq 2|E|$ — ①

and degree is at most $\Delta(G)$ $\therefore \Delta(G) \cdot |V| \geq 2|E|$ — ②

By ① & ②

$$\delta(G) \cdot |V| \leq 2|E| \leq \Delta(G) \cdot |V|$$

#Q. The number of edges in a k - regular graph with n vertices is

$$k \cdot |V| = 2|E|$$

$$k \cdot n = 2 \cdot |E|$$

$$|E| = \frac{nk}{2}$$

#Q. A non-directed graph contains 16 edges and all vertices are of degree 2. Then the number of vertices in G is ____.

$|E|$

$K=2$

$$K |V| = 2 |E|$$

$$2 * |V| = 2 * 16$$

$$|V| = 16$$

#Q. G is undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is _____.

at least 3

$$\hookrightarrow \therefore 3|V| \leq 2|E|$$

$$3 \cdot n \leq 2 \times 25$$

$$n \leq \frac{50}{3}$$

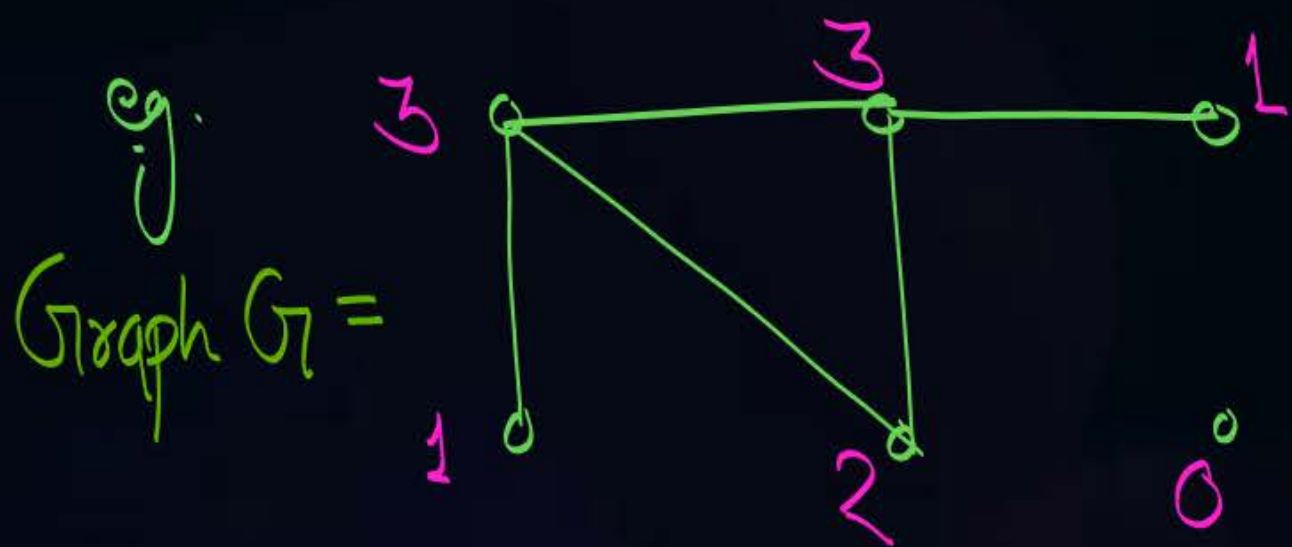
$$n \leq 16.66 \dots$$

$$\boxed{n \leq 16} \Rightarrow \underline{n_{\max} = 16}$$



Topic : Degree sequence

In a graph G if degrees of all the vertices are arranged in non-increasing or non-decreasing order, then it is called degree sequence of graph G .



Degree Sequence of graph G in non-increasing order $= (3, 3, 2, 1, 1, 0)$

Degree Sequence of graph G in non-decreasing order $= (0, 1, 1, 2, 3, 3)$

Graphic:- If there exists a simple non-directed graph w.r.t given degree sequence then that degree sequence is called a graphic, Converse of the statement is also true!

#Q. Which of the following degree sequences represent a simple non-directed graph? *{i.e. which of the following is/are graphic?}*

1. $\{2, 3, 3, 4, 4, 5\}$
2. $\{2, 3, 4, 4, 5\}$
3. $\{1, 3, 3, 3\}$
4. $\{0, 1, 2, 3, \dots, n-1\}$
5. $\{1, 3, 3, 4, 5, 6, 6\}$
6. $\{3, 3, 3, 3, 2\}$

#Q. Which of the following degree sequences represent a simple non-directed graph? {i.e. which of the following is/are graphic?}

1. $\{2, 3, 3, 4, 4, 5\}$ $\Rightarrow$ No. of vertices with odd degree can never be odd in any graph (simple + not-simple).
o's Given degree sequence does not represent any graph. Hence it is not "a graphic"
2. $\{2, 3, 4, 4, 5\}$
3. $\{1, 3, 3, 3\}$
4. $\{0, 1, 2, 3, \dots, n-1\}$
5. $\{1, 3, 3, 4, 5, 6, 6\}$
6. $\{3, 3, 3, 3, 2\}$

#Q. Which of the following degree sequences represent a simple non-directed graph?

1. {2, 3, 3, 4, 4, 5}

2. {2, 3, 4, 4, ~~5~~}

3. {1, 3, 3, 3}

4. {0, 1, 2, 3, ..., n-1}

5. {1, 3, 3, 4, 5, 6, 6}

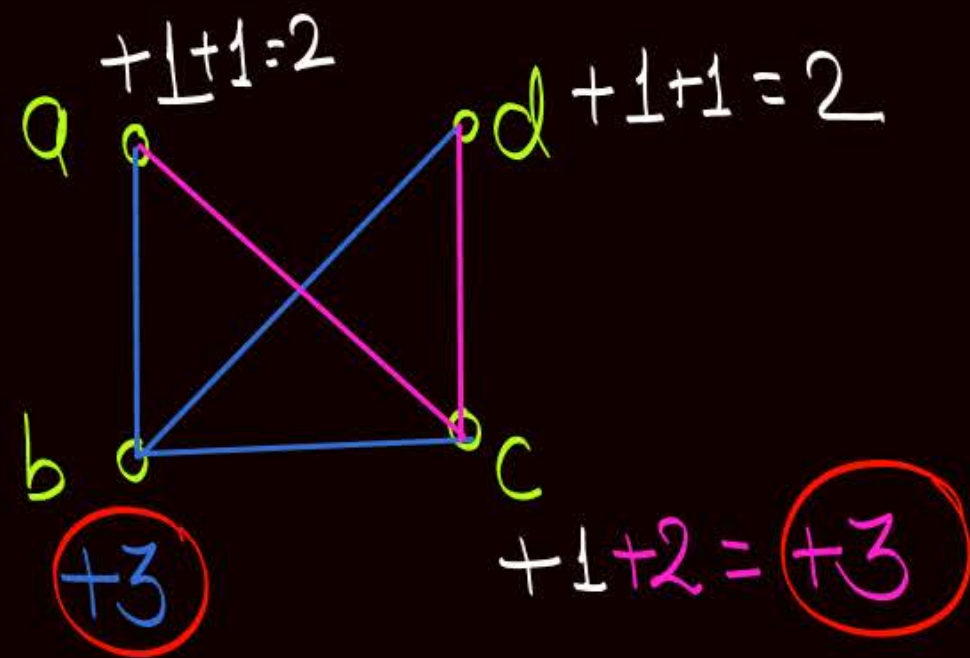
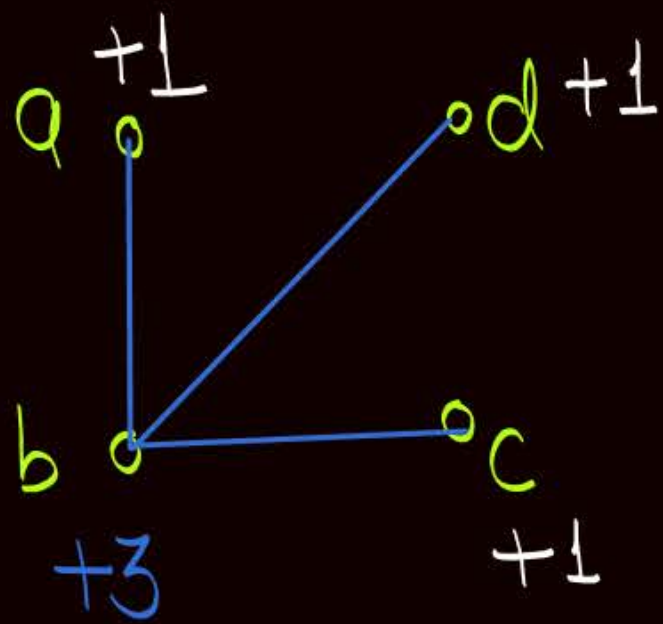
6. {3, 3, 3, 3, 2}

In a simple graph with n -vertices,
degree of every vertex $\leq (n-1)$

There are '5' degrees in the sequence
i.e., '5' vertices in the graph

↳ In a simple graph with '5' vertices
degree of a vertex can be at most '4'
(5-1)

↳ $\therefore$ Vertex of degree = 5
is not possible



no. of vertices " $n=4$ "
 i. Max degree possible
 for a vertex in a
 Simple graph = $(4-1) = 3$

Note:- In a simple graph with ' n ' vertices,
 if there are ' k ' vertices of degree = $(n-1)$,
 then in that simple graph degree
 of every vertex must be 'at least k '

#Q. Which of the following degree sequences represent a simple non-directed graph?

1. $\{2, 3, 3, 4, 4, 5\}$

2. $\{2, 3, 4, 4, 5\}$

3. ~~$\{1, 3, 3, 3\}$~~

4. $\{0, 1, 2, 3, \dots, n-1\}$

5. $\{1, 3, 3, 4, 5, 6, 6\}$

6. $\{3, 3, 3, 3, 2\}$

No. of vertices = 4

And there are '3' vertices of $\text{deg} = 3 \{ \text{i.e. } 4-1 \}$

↳ Hence in a simple graph degree of every vertex must be at least '3'

↳ And hence vertices of $\text{deg} = 1$ can not exist

#Q. Which of the following degree sequences represent a simple non-directed graph?

1. {2, 3, 3, 4, 4, 5}

2. {2, 3, 4, 4, 5}

3. {1, 3, 3, 3}

4. ~~{0, 1, 2, 3, ..., n-1}~~

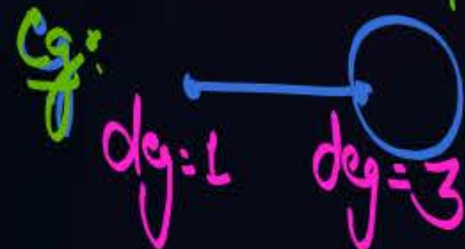
5. {1, 3, 3, 4, 5, 6, 6}

6. {3, 3, 3, 3, 2}

Total no. of vertices = 'n'
And there is '1' vertex of degree = (n-1)

∴ In a simple graph degree of every vertex must be at least '1'
Hence vertex of deg=0 can not exist

Not true for a graph which is not simple



Note: In a simple graph all vertices can not have distinct degrees.
i.e., In a simple graph there must be at least two vertices of same degree

#Q. Which of the following degree sequences represent a simple non-directed graph?

No. of vertices = 7
and we have '2' vertices of $\text{deg} = 6 = (7-1)$

∴ In a simple graph degree of every vertex must be at least '2'.

Hence in a graphic vertex of $\text{deg} = 1$ is not possible

1. {2, 3, 3, 4, 4, 5}

2. {2, 3, 4, 4, 5}

3. {1, 3, 3, 3}

4. {0, 1, 2, 3, ..., n-1}

5. ~~{1, 3, 3, 4, 5, 6, 6}~~

6. {3, 3, 3, 3, 2}

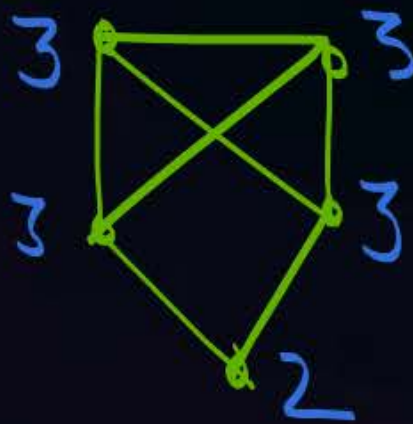
#Q. Which of the following degree sequences represent a simple non-directed graph?

1. $\{2, 3, 3, 4, 4, 5\}$
2. $\{2, 3, 4, 4, 5\}$
3. $\{1, 3, 3, 3\}$
4. $\{0, 1, 2, 3, \dots, n-1\}$
5. $\{1, 3, 3, 4, 5, 6, 6\}$
6. $\{3, 3, 3, 3, 2\}$

None of conditions based on which we can say that "it is not a graphic" is satisfied.

∴ We will have to check explicitly.

We can observe that there is a simple graph wrt. given degree sequence.



Hence given degree sequence is a graphic.

#Q. Which of the following degree sequences represent a simple non-directed graph? (or) Which of the following deg. sequences is/are graphic?

~~1.~~ {2, 3, 3, 4, 4, 5}

~~2.~~ {2, 3, 4, 4, 5}

~~3.~~ {1, 3, 3, 3}

~~4.~~ {0, 1, 2, 3, ..., n-1}

~~5.~~ {1, 3, 3, 4, 5, 6, 6}

✓ 6. {3, 3, 3, 3, 2}



Topic : Havel Hakimi's Result

It is used to check
whether the given degree
Sequence is a graphic
or not

"Next slide"

Havel Hakimi's Algorithm

Consider the following degree sequence in non-increasing order

Seq 1: $\{S, t_1, t_2, t_3, \dots, t_s, d_1, d_2, \dots, d_k\}$ $\{S, t_1, t_2, \dots, d_1, d_2, \dots \text{etc.}\}$
[are degrees of different vertices]

In order to reduce the size of the problem delete the first degree from the sequence (i.e., delete 'S'), and subtract '1' from the next 'S' degrees, and obtain a new deg seq "Seq 2"

Seq 2: $\{(t_1-1), (t_2-1), \dots, (t_s-1), d_1, d_2, \dots, d_k\}$

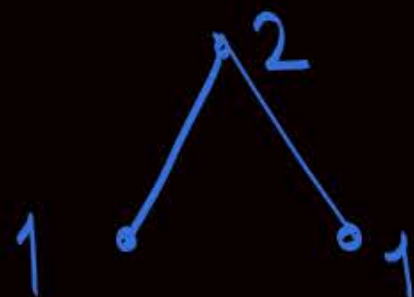
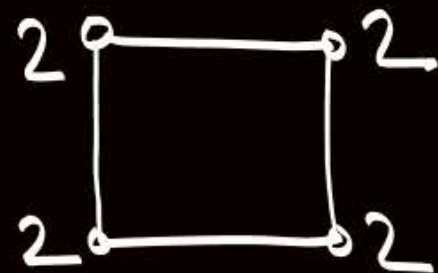
"Seq 1" Can represent a simple non-directed graph if and only if
"Seq 2" can represent a simple non-directed graph.

If we are not able to visualize the graph w.r.t. "Seq 2" as well, then rearrange the "Seq 2" in non-increasing order and repeat the process to obtain "Seq 3" {keep repeating until you are able to visualize the graph}

Q:- Check whether the following degree sequence can represent a simple non-directed graph or not?

Seq 1: $\{ \cancel{3}, \underbrace{3, 3, 3}, 2 \}$

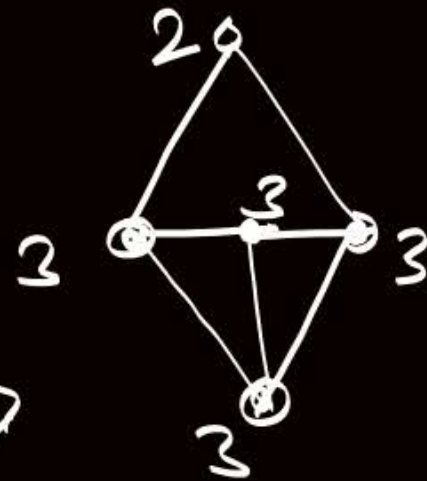
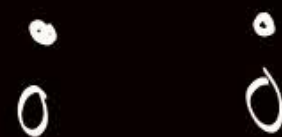
Seq 2: $\{ \cancel{2}, \underbrace{2, 2, 2}, 2 \}$



Seq 3: $\{ 1, 1, 2 \}$

Rearranged Seq 3: $\{ \cancel{2}, \underbrace{1, 1}, 1 \}$

Seq. 4: $\{ 0, 0 \}$



#Q. Which of the following degree sequences represent a simple non directed graph?

$$S1 = \{6, 6, 6, 6, 4, 3, 3, 0\}$$

$$S2 = \{6, 5, 5, 4, 3, 3, 2, 2, 2\}$$

- A** Only S1
- B** Only S2
- C** Both S1 and S2
- D** Neither S1 nor S2

#Q. Which of the following degree sequences represent a simple non directed graph?

$S_1 = \{6, 6, 6, 6, 4, 3, 3, 0\}$

$S_2 = \{6, 5, 5, 4, 3, 3, 2, 2, 2\}$

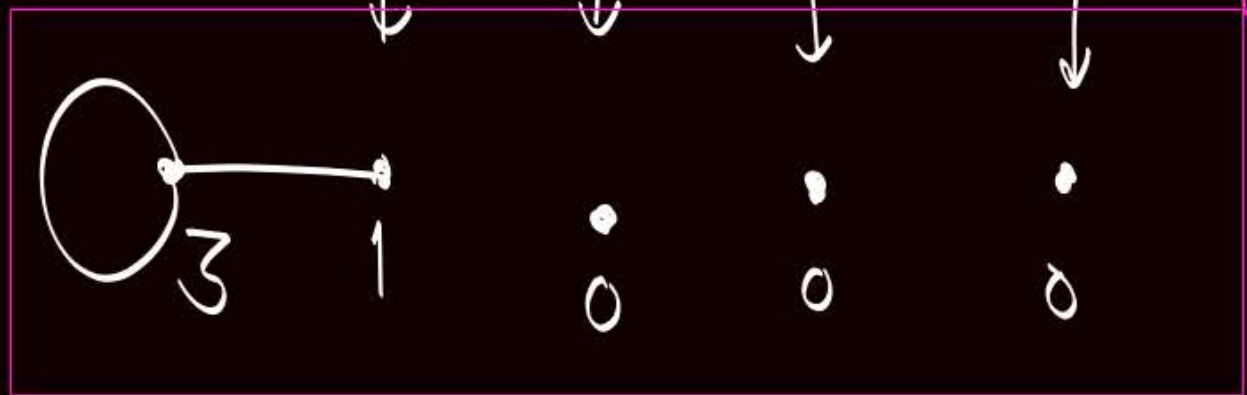
- A** Only S_1
- B** Only S_2
- C** Both S_1 and S_2
- D** Neither S_1 nor S_2

*∴ degree of Every other vertex must be at least '4'.
i.e. vertices of $\text{deg} = 3$ Can not exist*

• "ignore"

*4 vertices of
degree (7-1)*

$S_1 \Rightarrow \{6, 6, 6, 6, 4, 3, 3, 0\}$
 Seq 1: $\{\cancel{6}, 6, 6, 6, 4, 3, 3, 0\}$
 Seq 2: $\{\cancel{5}, 5, 5, 3, 2, 2, 0\}$
 Seq 3: $\{\cancel{4}, 4, 2, 1, 1, 0\}$
 Seq 4: $\{3, 1, 0, 0, 0\}$



final sequence
 Can not represent
 Any simple
 non-directed graph
 i.e. S_1 also Can not
 represent any
 simple graph.

$S_2 \Rightarrow \{6, 5, 5, 4, 3, 3, 2, 2, 2\}$

Seq 1: $\{\cancel{6}, 5, 5, 4, 3, 3, 2, 2, 2\}$

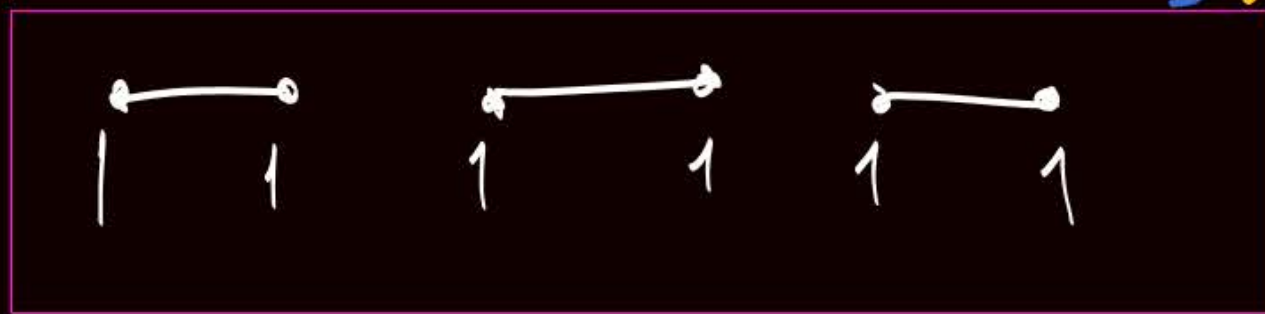
Seq 2: $\{4, 4, 3, 2, 2, 1, 2, 2\}$

Re arranged Seq 2: $\{\cancel{4}, 4, 3, 2, 2, 2, 2, 1\}$

Seq 3: $\{3, 2, 1, 1, 2, 2, 1\}$

Re arranged Seq 3: $\{\cancel{3}, 2, 2, 2, 1, 1, 1\}$

Seq 4: $\{1, 1, 1, 1, 1, 1\}$



Final Seq.
Can represent
A simple -
non-directed graph.
 $\therefore S_2$ also can
represent a
simple non-directed
graph

#Q. Which of the following degree sequences represent a simple non directed graph?

~~$S_1 = \{6, 6, 6, 6, 4, 3, 3, 0\}$~~

$S_2 = \{6, 5, 5, 4, 3, 3, 2, 2, 2\}$

A Only S_1

B Only S_2

C Both S_1 and S_2

D Neither S_1 nor S_2



Topic : Complement of a graph

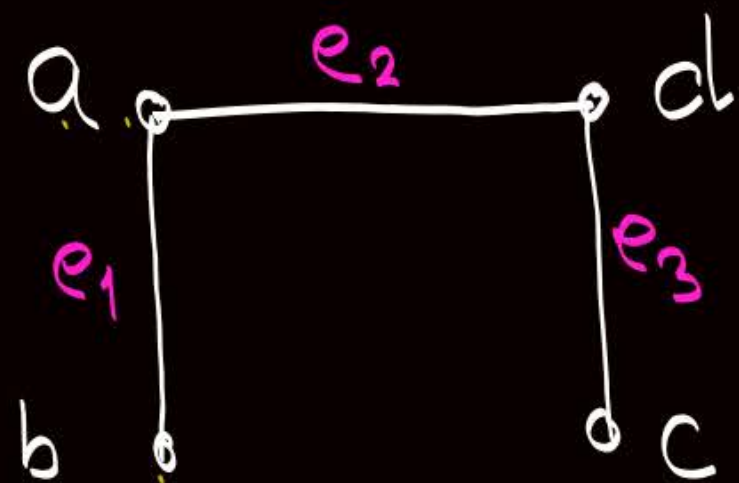


{ Complement of a graph can be defined if and only if graph is a simple graph }

- Let G be a simple graph with n -vertices, then complement of graph G is a simple graph with same n -vertices as of G but an edge is present in complement of graph G if and only if that edge is not present in G .

Complement of graph G can be denoted by $\overline{G}$

eg:



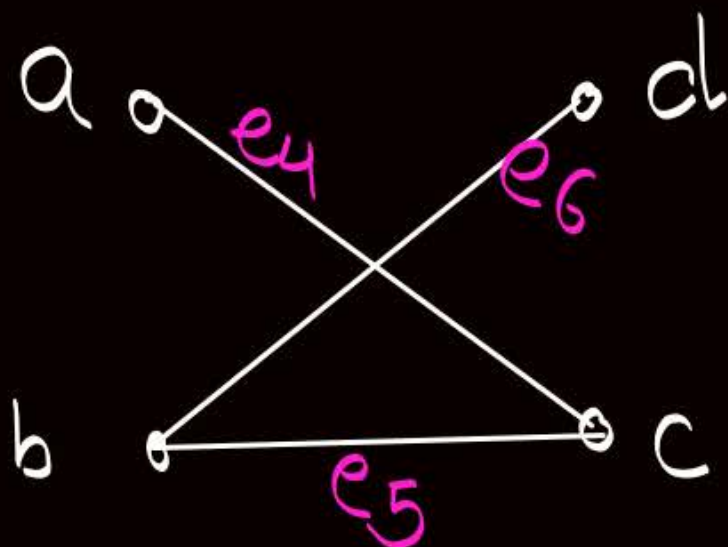
G

$$G = (V, E_1)$$

$$V = \{a, b, c, d\}$$

$$E_1 = \{e_1, e_2, e_3\}$$

$\cup$



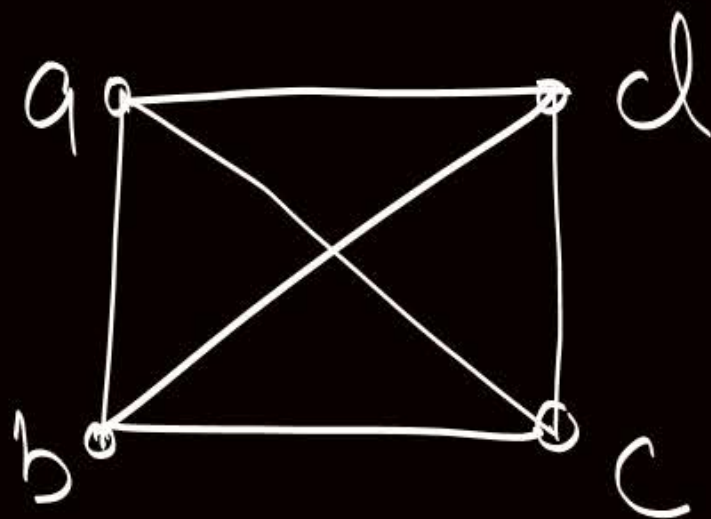
$\overline{G}$

$$\overline{G} = (V, E_2)$$

$$V = \{a, b, c, d\}$$

$$E_2 = \{e_4, e_5, e_6\}$$

$=$



$$G \cup \overline{G} = (K_4)$$

$$G \cup \overline{G} = (V \cup V, E_1 \cup E_2)$$

$$V \cup V = V = \{a, b, c, d\}$$

$$E_1 \cup E_2 = \{e_1, e_2, e_3, e_4, e_5, e_6\}$$

Note

→ Let G be a simple graph with n -vertices
and $\bar{G}$ is complement of graph G .

then,

$$G \cup \bar{G} = K_n$$

Note

$$|V(G)| = |V(\bar{G})|$$

&

$$|E(G)| + |E(\bar{G})| = |E(K_n)|$$

$$|E(G)| + |E(\bar{G})| = \frac{n(n-1)}{2}$$

Q: Let G be a simple graph with n -vertices and 21 edges.
If there are '24' edge in Complement of graph G ,
then find the value of ' n '.

Soln

$$|E(G)| + |E(\bar{G})| = \frac{n(n-1)}{2}$$

$$21 + 24 = \frac{n(n-1)}{2}$$

$$90 = n(n-1)$$

$$\therefore n = -9 \text{ or } 10$$

But no. of vertices can not be negative
 $\therefore$ $n = 10$

Note:- *

Let G be a simple graph with ' n ' vertices
and following degree sequence.

$$\text{degree seq. of } G = \{x_1, x_2, x_3, \dots, x_n\}$$

then,

$$\text{degree sequence of } \overline{G} = \{(n-1)-x_1, (n-1)-x_2, (n-1)-x_3, \dots, (n-1)-x_n\}$$

Note:-

Let G is a simple graph and $\bar{G}$ is Complement of graph G , then at least one of G or $\bar{G}$ is a Connected graph

↳ ∴ Both of them can never be disconnected,

{ But both of them may be }
Connected



2 mins Summary



✓
Topic

Sum of degree theorem

✓
Topic

Degree sequence

✓
Topic

Graphic

✓
Topic

Havel Hakimi's algorithm

Complement of a graph

Next class:-

- ① Graph isomorphism
- ② Self-Complementary graph
- ③ Planar graph

THANK - YOU